# Bellbeat Wellness Analysis with R

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## Introduction

This case study is part of the Google Data Analytics Capstone project and focuses on analyzing smart device usage data to generate actionable insights for a high-tech wellness company, **Bellabeat**. Bellabeat is a health-focused brand that manufactures smart wellness products such as activity trackers, and watches, which helps users monitor fitness, stress, and sleep.

As a recent Economics graduate from Christopher Newport University I found this choice out of the projects the most interesting. It gives me the ability to showcase my analytical skills that can hopefully lead to actionable decision making. I will be using the Fitbit fitness tracker data to explore patterns in user behavior related to physical activity, sleep, and energy expenditure. The dataset includes information on steps, distances, calories burned, sleep duration, and other daily health metrics.

## Installing packages and Importing Data

```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3

## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'stringr' was built under R version 4.4.3

## Warning: package 'forcats' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
```

```
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
daily_activity <- read.csv("C:/Users/Colin/OneDrive/Desktop/Google Capstone/mturkfitbit_export_3.12.16-4
sleep_day <- read.csv("C:/Users/Colin/OneDrive/Desktop/Google Capstone/mturkfitbit_export_4.12.16-5.12.1
```

```
n_distinct(daily_activity$Id)
```

```
## [1] 35
```

```
n_distinct(sleep_day$Id)
```

```
## [1] 24
```

## Data Overview

```
nrow(daily_activity)
```

```
## [1] 457
```

```
nrow(sleep_day)
```

```
## [1] 413
```

There are 457 daily activity records and 413 sleep records.

```
daily_activity %>%
  select(TotalSteps, TotalDistance, SedentaryMinutes, VeryActiveMinutes, Calories) %>%
  summary()
```

```
##    TotalSteps    TotalDistance    SedentaryMinutes VeryActiveMinutes
##  Min.   :    0   Min.   : 0.000   Min.   :  32.0   Min.   :  0.00
##  1st Qu.: 1988   1st Qu.: 1.410   1st Qu.: 728.0   1st Qu.:  0.00
##  Median : 5986   Median : 4.090   Median :1057.0   Median :  0.00
##  Mean   : 6547   Mean   : 4.664   Mean   : 995.3   Mean   : 16.62
##  3rd Qu.:10198   3rd Qu.: 7.160   3rd Qu.:1285.0   3rd Qu.: 25.00
##  Max.   :28497   Max.   :27.530   Max.   :1440.0   Max.   :202.00
##     Calories
##  Min.   :   0
##  1st Qu.:1776
##  Median :2062
##  Mean   :2189
##  3rd Qu.:2667
##  Max.   :4562
```

```r
sleep_day %>%
  select(TotalSleepRecords,
         TotalMinutesAsleep,
         TotalTimeInBed) %>%
  summary(sleep_day)
```

```
##  TotalSleepRecords TotalMinutesAsleep TotalTimeInBed
##  Min.   :1.000     Min.   : 58.0      Min.   : 61.0
##  1st Qu.:1.000     1st Qu.:361.0      1st Qu.:403.0
##  Median :1.000     Median :433.0      Median :463.0
##  Mean   :1.119     Mean   :419.5      Mean   :458.6
##  3rd Qu.:1.000     3rd Qu.:490.0      3rd Qu.:526.0
##  Max.   :3.000     Max.   :796.0      Max.   :961.0
```

## Observation From Summaries

Users averaged 6,547 steps, 2,189 calories burned, and 16.62 very active minutes.

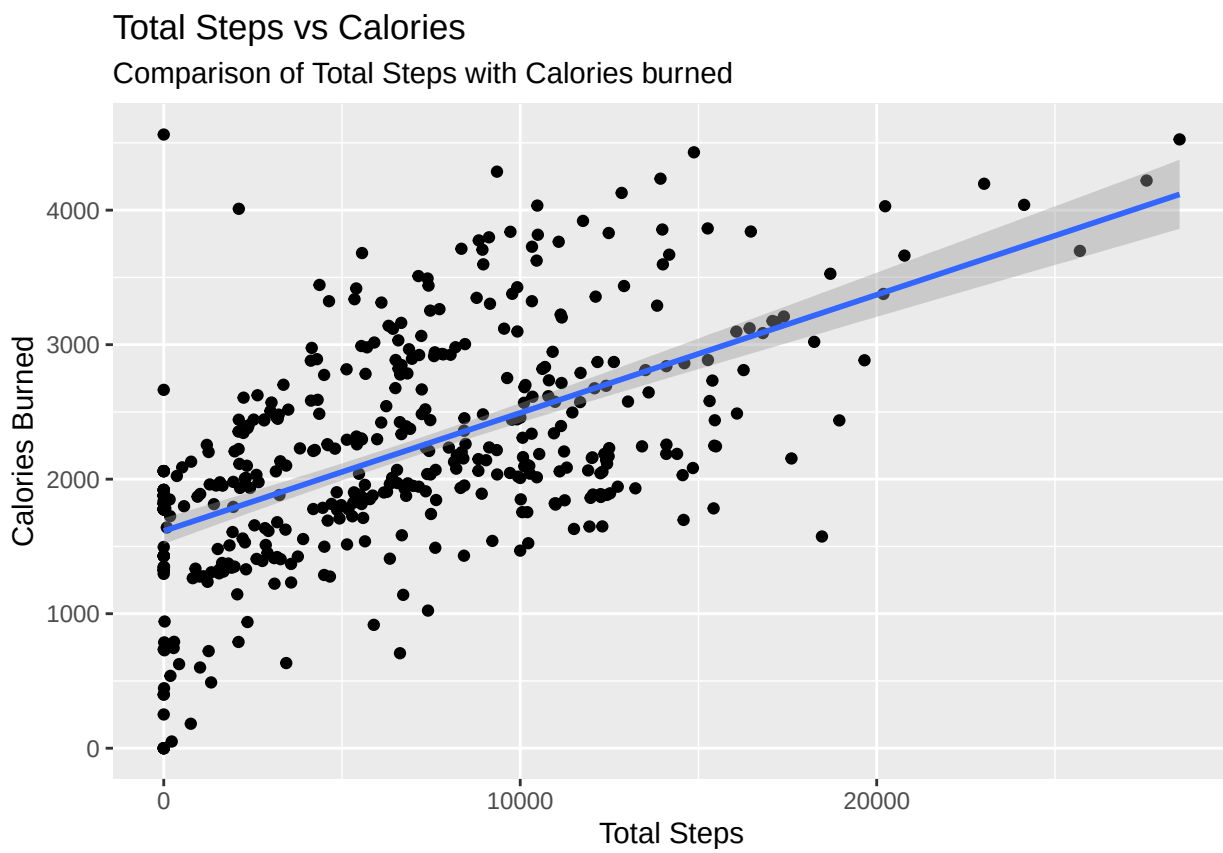
On average, people spent about 40 extra minutes in bed not sleeping, indicating potential for improving sleep quality.

Most users were sedentary for 16.5 hours or 995 minutes per day, major opportunity for movement coaching.

Visualizations

```r
ggplot(data = daily_activity, aes(x = TotalSteps, y = Calories)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    title = "Total Steps vs Calories",
    subtitle = "Comparison of Total Steps with Calories burned",
    x = "Total Steps",
    y = "Calories Burned"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




### Viz Observation(1)

From this observation there is a positive correlation between the amount of total steps taken and calories burned. This shows that having a higher amount of steps taken a day will result in more calories burned.

```
ggplot(data = daily_activity, aes(x = TrackerDistance, y = Calories)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    title = "Tracker Distance vs Calories",
    subtitle = "Calories burned over a distance",
    x = "Tracked Distance (miles)",
    y = "Calories Burned"
  )
```

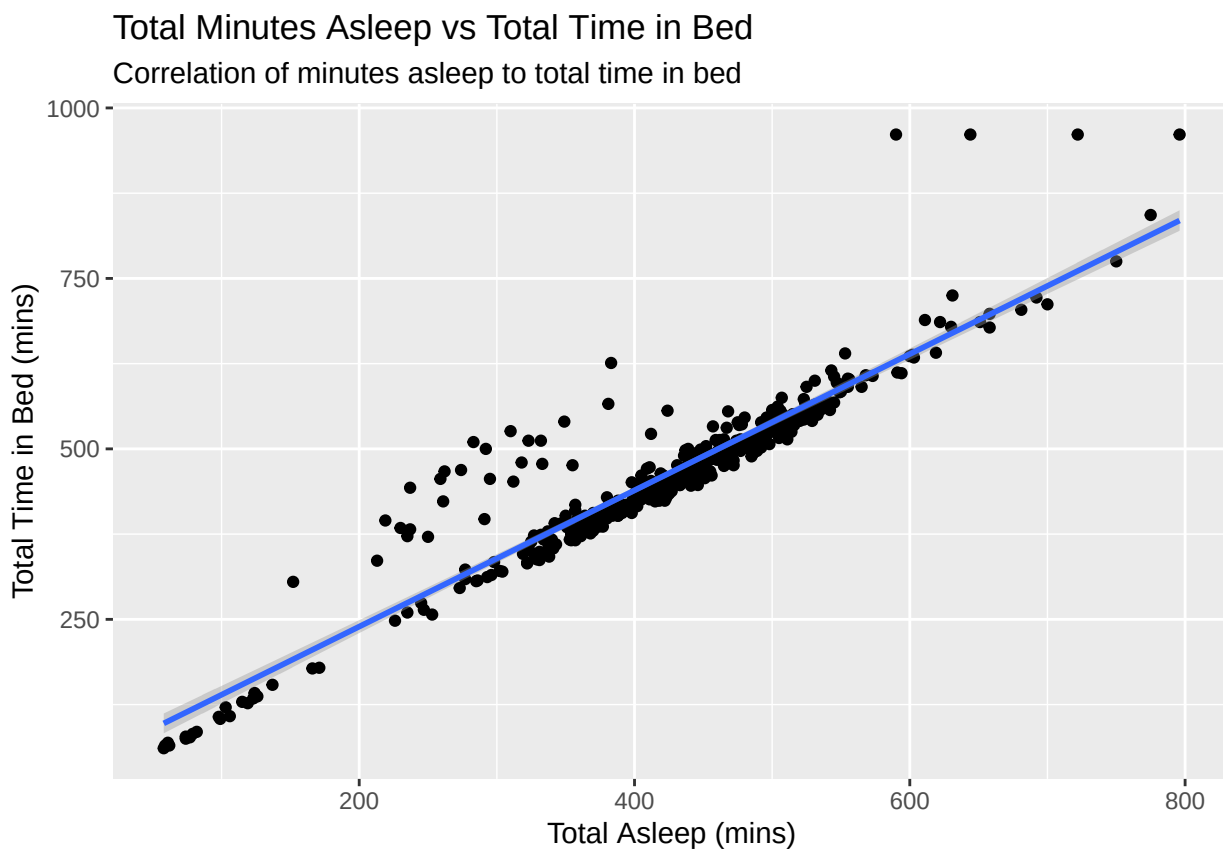
```
## `geom_smooth()` using formula = 'y ~ x'
```

### Viz Observation(2)

A strong positive correlation shows that more distance walked leads to more calories burned expected and supports promoting activity tracking.

```
ggplot(data = sleep_day, aes(x = TotalMinutesAsleep, y = TotalTimeInBed)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    title = "Total Minutes Asleep vs Total Time in Bed",
    subtitle = "Correlation of minutes asleep to total time in bed",
    x = "Total Asleep (mins)",
    y = "Total Time in Bed (mins)"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




### Viz Obseravtion(3)

There is a strong linear relationship between minutes asleep and time in bed. However, some users consistently spend more time in bed than asleep, signaling possible inefficiencies.

### Merging Data

```
combined_data <- merge(sleep_day, daily_activity, by = "Id")
n_distinct(combined_data$Id)
```

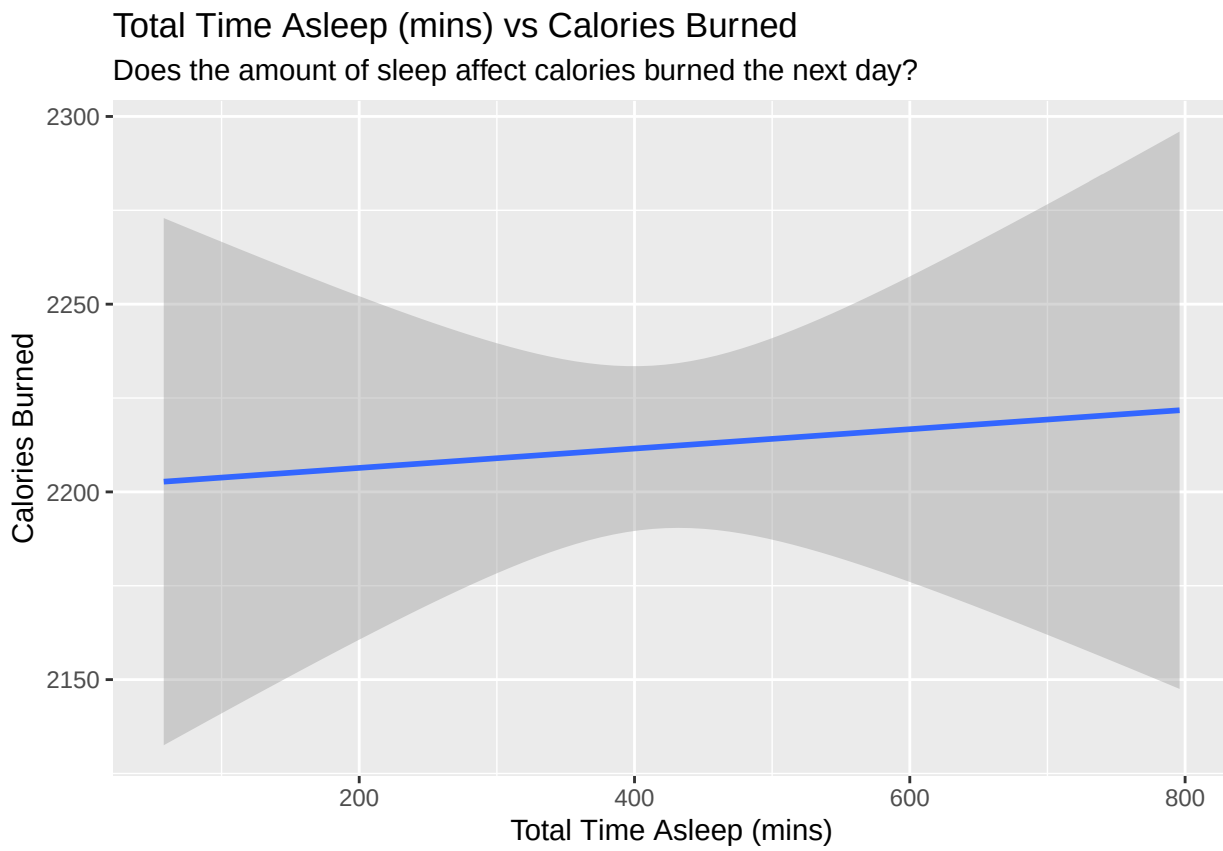
```
## [1] 24
```

### Merged Data Visualizations

```
ggplot(data = combined_data, aes(x = TotalMinutesAsleep, y = Calories)) +
  geom_smooth(method = "lm") +
  labs(
    title = "Total Time Asleep (mins) vs Calories Burned",
    subtitle = "Does the amount of sleep affect calories burned the next day?",
    x = "Total Time Asleep (mins)",
```

```
    y = "Calories Burned"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

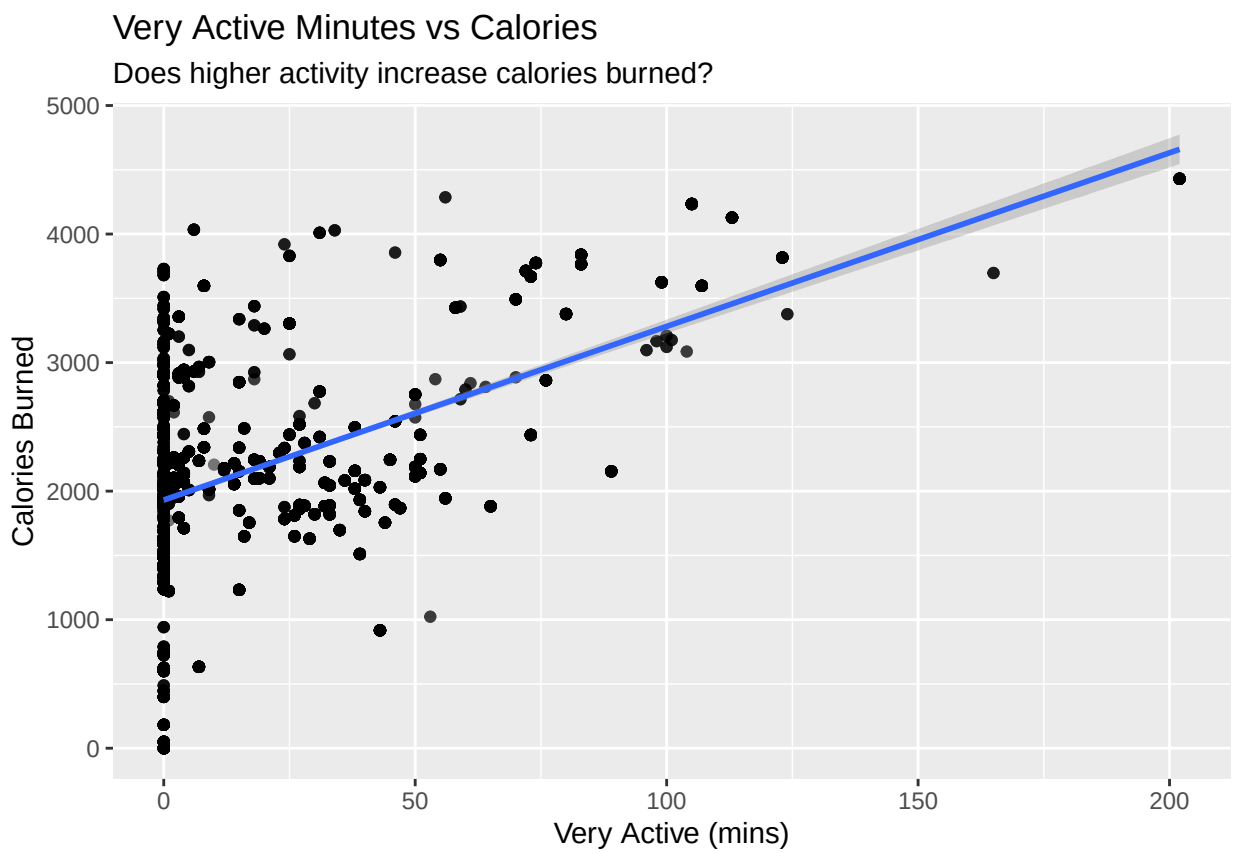



### Viz obseravtion(4)

There is no significant correlation between sleep duration and calories burned. This suggests sleep does not directly impact calories, though it may influence activity energy indirectly.

```
ggplot(data = combined_data, aes(x = VeryActiveMinutes, y = Calories)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm") +
  labs(
    title = "Very Active Minutes vs Calories",
    subtitle = "Does higher activity increase calories burned?",
    x = "Very Active (mins)",
    y = "Calories Burned"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




## Viz Obseravtion (5)

There is a strong positive correlation higher very active minutes consistently result in more calories burned. This is one of the most predictive activity measures in the dataset.

## Conclusion & future decsions.

### Key Findings

Activity level (steps, distance, and very active minutes) is strongly linked to calories burned.

Sleep duration is not directly related to calories burned but may influence general wellness.

Many users are sedentary for long periods 16.5 hours a day on average.

Sleep inefficiencies (extra time in bed) may present opportunities for coaching.

## Recommendations for Bellabeat

### Promote activity tracking features:

Market very active minutes and calorie burn stats.

Motivate users to break up long sedentary periods.

**Target segments with low movement:**

Encourage small daily goals (steps, distance).

Use notifications or challenges to encourage breaks and walks.

**Build a focus on sleep quality:**

Educate users on reducing time in bed while awake.

Introduce guided bedtime routines or relaxation features in the app.

**Use segmentation for personalized messaging:**

High activity users: promote performance stats and fitness goals.

Low activity users: focus on behavior change and building habits.